TASK1: I selected the GradientBoostingClassifier because Kickstarter outcomes depend on nonlinear interactions among project attributes such as goal size, name length, category, and country. To prepare the dataset for modeling, I focused on retaining only information available before launch. This involved removing leakage variables (e.g. pledge), redundant timestamp components(e.g. hour), and identifiers that do not contribute predictive information (e.g. name). Next, I restricted the dataset to rows where the project state is either 'successful' or 'failed'. To pre-process, I begin by checking duplicated rows, unary variables, and missing values. Based on the similarity of the main_category and category, I choose to use category to fill the missing value in main_category. However, considering the multicollinearity, I removed currency, category and redundant text-length variables after testing the internal correlation. Time variables were transformed into meaningful durations (lead time and durations), and I created a weekend-launch indicator to capture potential timing effects. I also examined the distribution of the country variable and found many very low-frequency categories. I kept countries with at least 50 observations and grouped the rest into an "Other" category. Goal amounts were converted to USD and log-transformed to address skewness. All categorical variables were one-hot encoded and boolean values were converted into numbers. The final target variable is "y = state", and the final feature matrix X is all columns in df except "state". **Model:** I split the data into training (70%) and testing (30%) sets and standardized the predictors using "StandardScaler", fitting on the training set only to prevent data leakage. Next, I do a feature selection by Lasso (alpha = 0.005), and build a GradientBoostingClassifier Grid search to tune hyperparameters, having a final accuracy score of 76%. With our pre-processing steps and Lasso feature selection, our model has a robust predicting ability of its outcomes at the beginning of time each project is launched.

In a business context, our classification model provides actionable insights for both Kickstarter and project creators by enabling data-driven decision-making. The model predicts a campaign's likelihood of success at launch, allowing the platform to intervene in meaningful ways. Low-probability projects can be flagged early so that creators receive targeted recommendations, such as adjusting their funding goal, improving media content, or refining project descriptions. The predictions also help Kickstarter allocate limited promotional resources, such as Staff Picks and visibility, to campaigns that are more likely to convert that exposure into actual pledges. Beyond individual campaigns, the model supports platform-level planning by offering early forecasts on funding volume, category performance, and creator behavior trends. Overall, the model enhances decision-making for both creators and the platform, improving campaign outcomes and helping Kickstarter maintain a healthier, more successful ecosystem.

Task2: I used the K-Means algorithm because it offers an interpretable, geometry-based partition of the projects, allowing us to summarize Kickstarter campaigns into a manageable set of profiles. Before clustering, I do the same data mining steps except a few adjustments. I set the threshold of major countries to be 150 to reduce the dummy variable, and I delete the nearly-unary variable by a ratio of 0.98, and add back the after launching information. I experimented with different values of k and evaluated the results using Pseudo-F scores as it shows the optimal k value, and I use elbow graph method to validate the elbow point. Although silhouette values remained small, they stabilized at k = 4, providing a balance between interpretability and cluster cohesion.

Insights: Cluster 0 (865 projects) is the smallest group with moderate success rate, preparation time, and engagement signals. This segment has the lowest goal representing straightforward, low-visibility campaigns, often in category Crafts. Cluster 1 (1,615 projects)

consists of category-driven campaigns, particularly in Music, where outcomes rely heavily on niche audiences rather than broad platform exposure. Their longer preparation times and moderate engagement reflect a creator segment that is committed but audience-constrained. Cluster 2 (6,737 projects) is the largest group, with high-engagement, high-visibility, and popular campaigns. These projects invest heavily in preparation and consistently attract backers beyond their initial goals, making them strong candidates for platform-level amplification. Cluster 3 (5,246 projects) forms the second-largest segment. This group displays weaker presentation signals and generally lower expected engagement but high ambitions. The concentration of projects in this cluster highlights a systemic challenge on Kickstarter: many creators launch with high aspirations but insufficient campaign quality or strategic preparation.Beyond descriptive segmentation, the clustering model is useful for Kickstarter because it uncovers structural patterns in how campaigns are designed, prepared, and positioned—patterns that the platform cannot observe from success outcomes alone. Instead of predicting whether a project will succeed, clustering reveals how different types of campaigns behave before outcomes are known. This enables Kickstarter to tailor creator support strategies to the needs of each segment. For example, High-potential campaigns in Cluster 2 can be prioritized for promotional placement, recommendation engine boosts, or strategic partnerships. Creators in Cluster 3, who frequently struggle with campaign polish, can be proactively targeted with onboarding tutorials, automated content-quality diagnostics, or suggested goal adjustments. Meanwhile, niche creators in Clusters 0 and 1 may benefit from tailored audience-expansion tools, such as cross-category recommendations or micro-community outreach programs. Overall, the segmentation provides Kickstarter with a data-driven framework to optimize resource allocation, strengthen creator support workflows, and improve the platform’s long-term campaign success system.